AI Solutions Architect

Business Area: Professional Services

Seniority Level: Mid-Senior level

Job Description:

At Cloudera, we empower people to transform complex data into clear and actionable insights. With as much data under management as the hyperscalers, we're the preferred data partner for the top companies in almost every industry. Powered by the relentless innovation of the open source community, Cloudera advances digital transformation for the world’s largest enterprises.

As an AI Solutions Engineer within Cloudera’s Public Sector Consulting team, you will be the technical architect and execution lead for agencies moving from "data chaos" to "agentic autonomy." You will work directly with government organizations to design, build, and deploy mission-critical AI applications on the Cloudera Data Platform (CDP).

This is not a "theoretical" role. You will be on the front lines of Phase 2 and Phase 3 adoption journeys—helping customers clean legacy data silos, select the right model architectures, and industrialize MLOps pipelines in highly secure, often air-gapped or hybrid-cloud environments.

**As the AI Solutions Engineer you will:

1. AI Model Strategy, Selection and Implementation

Evaluate and select optimal model architectures (LLMs, SLMs, or traditional ML) based on mission requirements, considering tradeoffs between accuracy, latency, and cost.
Guide customers on "Build vs. Buy vs. Fine-tune" decisions, prioritizing open-source models (Llama, Mistral, Falcon) that can run securely within a sovereign data perimeter.
Experience building Agentic Workflows (AI agents that can execute API calls and multi-step tasks).
2. End-to-End Data Engineering

Design and implement robust data pipelines within CDP to transform "messy" legacy data into AI-ready formats.
Develop and optimize Vector Databases and Retrieval-Augmented Generation (RAG) architectures to ground AI responses in verified agency facts.
Build Data pipelines with Spark, Nifi, Kafka or other ETL tools.
3. Optimization & Performance Tuning

Optimize model inference for production environments using quantization, pruning, and hardware acceleration (NVIDIA GPU orchestration).

Implement LLMOps to monitor model performance, detect hallucination rates, and manage model versioning and drift.
4. Public Sector Advisory & Governance

Collaborate with the customer's AI Center of Excellence (CoE) to establish automated guardrails for ethics, bias mitigation, and FedRAMP/IL5 compliance.
Translate complex technical AI concepts into mission-value briefings for GS-level stakeholders and agency leadership.
**We're excited about you if you have: (Minimum Qualifications):

Experience: 5+ years in Data Engineering, Machine Learning, or Software Engineering, with at least 2 years focused on Generative AI or Deep Learning.
Technical Stack: Expertise in Python and deep learning frameworks (PyTorch, TensorFlow, Hugging Face).
Hands-on experience with Cloudera (CDP), Spark, or similar big data ecosystems.
Proficiency in orchestration tools like LangChain, LlamaIndex, or Haystack.
Experience developing visual data representations and dashboards (Django, React, or Angular)
Experience using a compiled programming language, preferably one that runs on the JVM (Java, Scala, etc)
Data Expertise: Proven ability to build ETL/ELT pipelines and work with both SQL and NoSQL/Vector databases (e.g., Pinecone, Milvus, or PGVector).
Public Sector Knowledge: Understanding of government security frameworks (NIST AI RMF, FedRAMP, SRGs, STIGs).
Active Top Secret Security Clearance
**You may also have: (Preferred Qualifications)

Experience fine-tuning of foundational models using techniques such as PEFT (Parameter-Efficient Fine-Tuning) and LoRA to adapt AI to domain-specific government nomenclature.
Experience training of specialized models on proprietary datasets while ensuring strict adherence to data privacy and sensitivity labels.
Experience installing and operating Cloudera Data Platform
Experience installing and operating Kubernetes
Experience in Air-Gapped deployments and managing AI workloads in disconnected environments.
Advanced degree (MS or PhD) in Computer Science, Data Science, or a related field.
Active Counterintelligence (CI) or Full Scope (FS) Poly is required.
This role is not eligible for immigration sponsorship.

What you can expect from us:

Generous PTO Policy
Support work life balance with Unplugged Days
Flexible WFH Policy

Mental & Physical Wellness programs
Phone and Internet Reimbursement program
Access to Continued Career Development
Comprehensive Benefits and Competitive Packages
Paid Volunteer Time
Employee Resource Groups